Week 02: Research Methods in Computer Science

R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

A proposal is only as convincing as its method. This week surveys the research methods used in computing, from controlled experiments and benchmarks to case studies, surveys, design science and formal analysis. It shows how to plan experiments with adequate power and how to report them reproducibly [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to match research questions with suitable research strategies and to identify threats to validity. They should be able to compute the sample size needed for a two-group comparison. They should also apply reproducibility practices when reporting computational experiments.

Lecture

The science of computing

Computer science combines mathematics, engineering and empirical science, and its methods reflect this mixture. Denning argued that computing is a science of information processes, natural and artificial, and that it follows the scientific method when it formulates and tests hypotheses [4]. Tedre and Moisseinen surveyed how the word experiment is used in computing and found several distinct meanings, from demonstrations of feasibility and trials of systems to comparisons and controlled experiments [5]. A proposal must therefore say precisely which kind of evidence it will produce.

Choosing a research strategy

Stol and Fitzgerald organised research strategies in software engineering by three desirable qualities that no single strategy can maximise at once [1]. The qualities are generalisability over actors, precision of measurement of behaviour and realism of context. Laboratory experiments

offer precision but limited realism, field studies offer realism but limited control, and sample studies such as surveys offer generalisability. Formal analyses and simulations complement them. Design science, as described by Hevner and colleagues, is a further strategy for building and evaluating artefacts that solve relevant problems, with guidelines on rigour, evaluation and communication [3]. Many computing proposals combine strategies, for example a design science cycle with a controlled benchmark and a field evaluation.

Experiments and validity

Wohlin and colleagues described the design of experiments in software engineering, from hypotheses and variables to the analysis of results, and classified threats to validity into four groups [2]. Conclusion validity concerns the statistical relation between treatment and outcome, and internal validity concerns whether the relation is causal. Construct validity concerns whether the measures capture the intended concepts, and external validity concerns generalisation. A frequent weakness of proposals is an experiment too small to detect the expected effect. Statistical power is the probability of detecting an effect of a given size if it exists. Figure 2.1 shows how power grows with the number of observations for small, medium and large effects. The interactive lab computes the sample size needed for a chosen effect size, significance level and power.

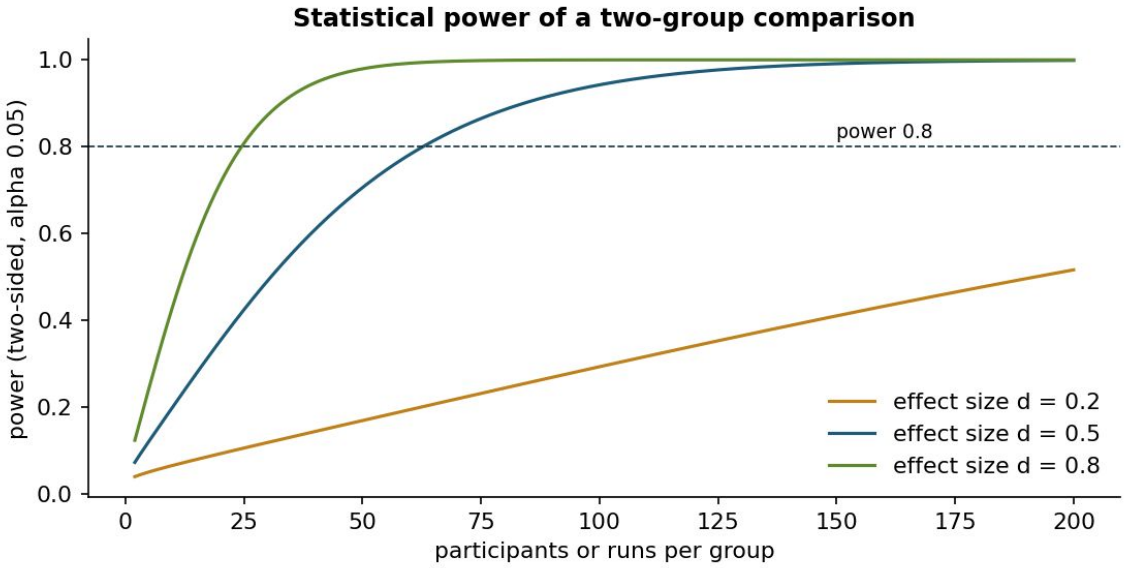


Figure 2.1. Power of a two-sided two-group comparison at a significance level of 0.05 as a function of group size for three effect sizes, computed with a normal approximation.

Reproducibility

Gundersen and Kjensmo examined papers from major artificial intelligence conferences and found that few documented their methods, data and experiments completely enough for independent reproduction [6]. Pineau and colleagues reported on the reproducibility programme of the NeurIPS 2019 conference, which combined a code submission policy, a reproducibility challenge and a checklist for authors [7]. Funders increasingly expect such practices to be

planned from the start. A method section that states the data, the code release, the random seeds, the statistical tests and the stopping rules is both more reproducible and more convincing to reviewers.

> ***Pause and reflect.*** *Which of the four validity threats is the greatest risk for the evaluation you plan in your project?*

Hands-on components

Interactive lab: Part A matches research questions with strategies. Part B sorts threats to validity into the four groups of Wohlin and colleagues. Part C computes the sample size per group for a two-group comparison [1, 2]. Link: https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-02/lab.html

Colab notebook: It simulates a benchmark comparison of two algorithms over many datasets, applies paired tests and effect sizes, and computes the sample size needed for a planned experiment [2]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-02/NB02_research_methods.ipynb

Reflection prompts

1. Which strategy fits your project's main question, and what do you give up by choosing it?
2. Describe one threat to validity in a paper you know well and how the authors could have addressed it.
3. What would a reviewer need in order to reproduce your planned experiments?

Weekly task and submission

What to do. Write the method section of your project, about 600 words. It covers six points: the research questions, the research strategy, the design of the study, the sample size with a power calculation, the main threats to validity and the plan for reproducibility.

Where to look. Lecture: sections "Choosing a research strategy", "Experiments and validity" and "Reproducibility". Interactive lab: Part A for strategies, Part B for threats to validity and Part C for the sample size. Colab notebook: Section 1 for a paired benchmark comparison and Section 2 for planning the sample size.

Example. A project that compares two code completion tools could run a controlled experiment with programmers as participants. For a large effect (d = 0.8), alpha = 0.05 and power 0.8, the formula of the Colab notebook gives about 25 participants per group. A threat to construct validity is measuring productivity only by the number of accepted suggestions.

What to submit. About 600 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Stol, K.-J., & Fitzgerald, B. (2018). The ABC of software engineering research. ACM Transactions on Software Engineering and Methodology, 27(3), 11. https://doi.org/10.1145/3241743

[2] Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in Software Engineering. Springer. https://doi.org/10.1007/978-3-642-29044-2

[3] Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. MIS Quarterly, 28(1), 75-105. https://doi.org/10.2307/25148625

[4] Denning, P. J. (2005). Is computer science science?. Communications of the ACM, 48(4), 27-31. https://doi.org/10.1145/1053291.1053309

[5] Tedre, M., & Moisseinen, N. (2014). Experiments in computing: A survey. The Scientific World Journal, 2014, 549398. https://doi.org/10.1155/2014/549398

[6] Gundersen, O. E., & Kjensmo, S. (2018). State of the art: Reproducibility in artificial intelligence. In Proceedings of the AAAI Conference on Artificial Intelligence. https://doi.org/10.1609/aaai.v32i1.11503

[7] Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d'Alché-Buc, F., Fox, E., & Larochelle, H. (2021). Improving reproducibility in machine learning research (A report from the NeurIPS 2019 reproducibility program). Journal of Machine Learning Research, 22(164), 1-20.